

TEST ON AUTOCLAVED AERATED BLOCK (IS 2185 :1984)

lightweight, precast, eco-friendly building materials made from cement, sand, lime, and fly ash, cured in an autoclave for high strength. They are roughly 50% lighter than traditional bricks, providing excellent thermal/sound insulation, faster construction, and significant structural.

These test preformed to check strength, durability, density, and insulation properties

- **Compressive Strength Test**
- **Dry Density Test**
- **Water Absorption Test**
- **Moisture Content Test**
- **Thermal Conductivity Test**
- **Drying Shrinkage Test**



Compressive Strength Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **compressive strength (load-bearing capacity)** of AAC blocks used in construction.

2. Apparatus

- **Compression Testing Machine (CTM)**
- **Steel scale**
- **Vernier caliper**
- **Weighing balance**

3. Procedure

1. Take a **clean and dry AAC block specimen**.
2. Measure the **length, width, and height** of the block.
3. Place the block **centrally in the Compression Testing Machine**.
4. Apply the **load gradually and uniformly** until the block fails or cracks.
5. Record the **maximum load at failure**.

4. Calculation

Compressive Strength = Maximum Load / Loaded Area

Compressive Strength = Loaded Area / Maximum Load

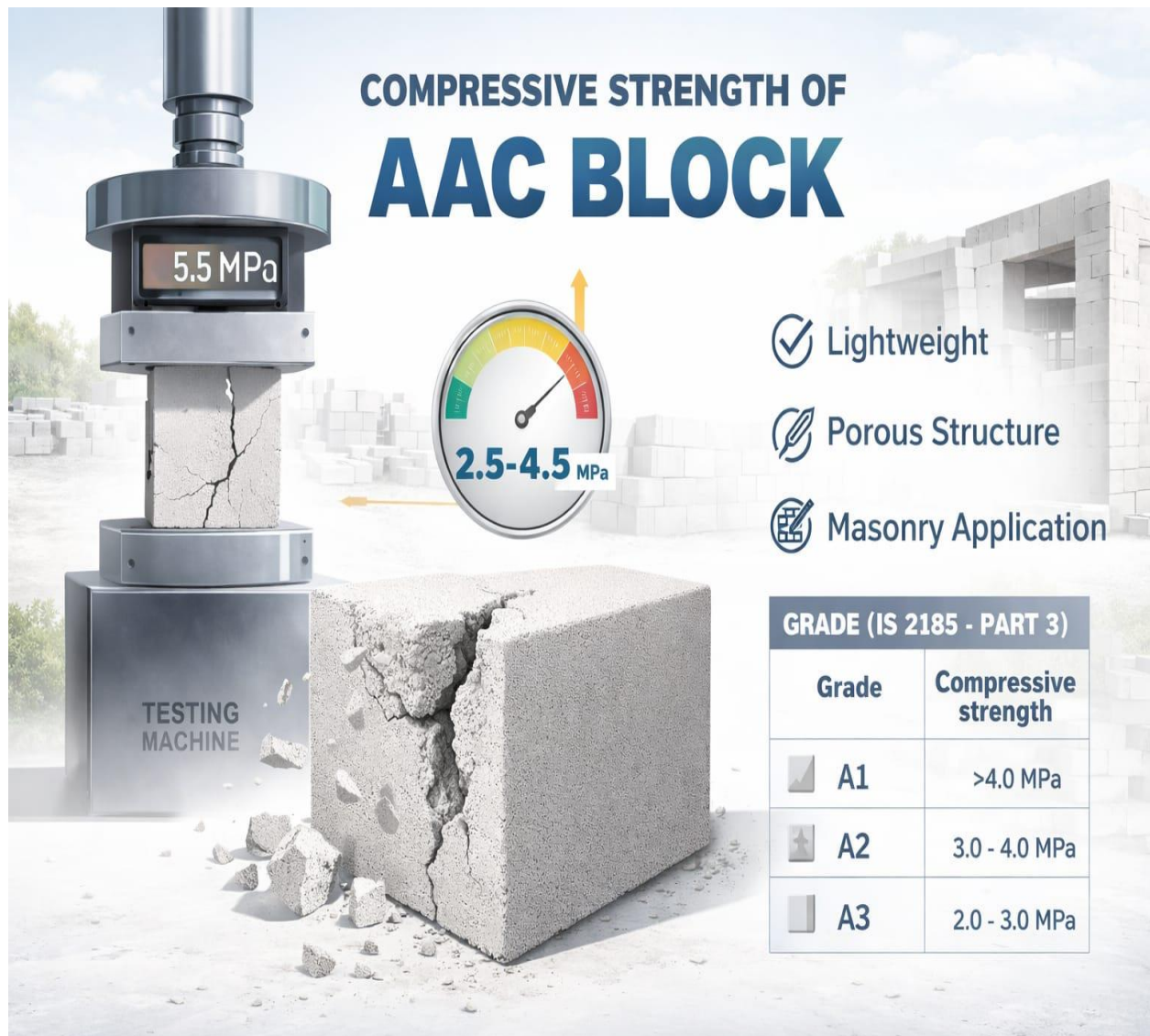
Unit: N/mm² (MPa)

5. Specifications

- Minimum compressive strength of AAC blocks: **3 – 4 N/mm²**

6. Result

The **compressive strength of the AAC block** is calculated and compared with the **standard value** to determine whether the block is suitable for construction.



Dry Density Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **dry density (mass per unit volume)** of AAC blocks used in construction.

2. Apparatus

- Weighing balance
- Vernier caliper / measuring scale
- Drying oven
- Tray

3. Procedure

1. Take a **clean AAC block specimen**.
2. Place the block in an **oven at about 105–110 °C** until it reaches constant weight.
3. Remove the block and allow it to **cool at room temperature**.
4. Measure the **length, width, and height** of the block.
5. Weigh the **dry block** using a weighing balance.

4. Calculation

Dry Density = Dry Weight of Block / Volume of Block

Block Volume = Length × Width × Height

Where

Volume = Length × Width × Height

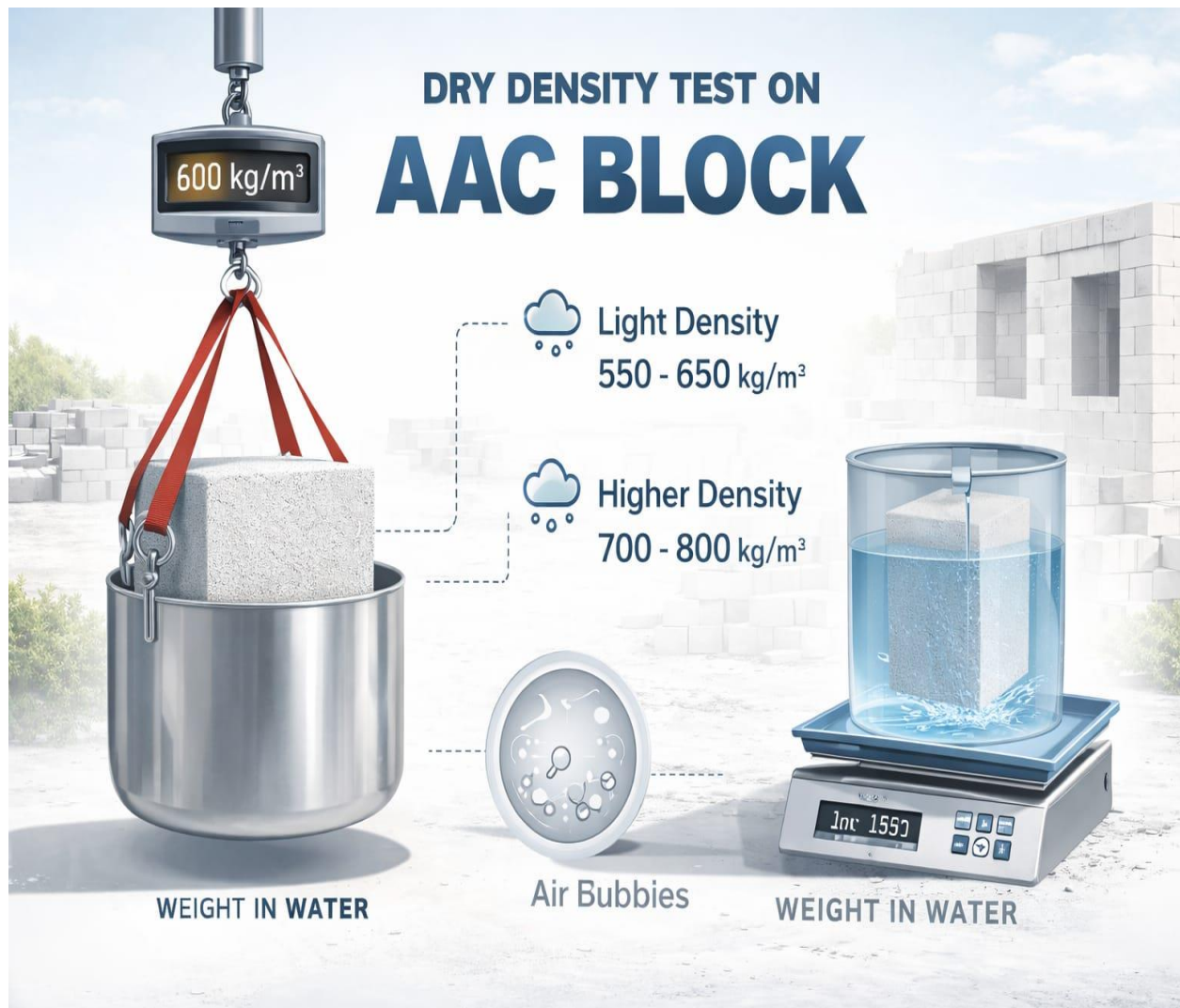
Unit: kg/m³

5. Specifications

- Typical dry density of AAC blocks: **550 – 650 kg/m³**

6. Result

The **calculated dry density** is compared with the **standard range** to check the quality and suitability of the AAC block for construction.



Water Absorption Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **amount of water absorbed by AAC blocks** when immersed in water.

2. Apparatus

- Weighing balance
- Water tank / container
- Dry cloth
- Oven (for drying)

3. Procedure

1. Take a **clean AAC block specimen**.
2. Dry the block in an **oven at about 105–110 °C** until constant weight is achieved.
3. Record the **dry weight (W_1)** of the block.
4. Immerse the block completely in **clean water for 24 hours**.
5. Remove the block, wipe off surface water with a **dry cloth**.
6. Record the **wet weight (W_2)** of the block.

4. Calculation

$$\text{Water Absorption (\%)} = \frac{W_2 - W_1}{W_1} \times 100$$

Where:

- **W_1 = Dry weight of block**
- **W_2 = Wet weight after immersion**

5. Specifications

- Water absorption of AAC blocks should generally be **not more than 10–15% by weight**.

6. Result

The **percentage of water absorption** is calculated and compared with the **standard limit** to determine the suitability of AAC blocks for construction.



Moisture Content Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **amount of moisture present in AAC blocks** before they are used in construction.

2. Apparatus

- Weighing balance
- Drying oven (105–110 °C)
- Tray or container
- Vernier caliper / measuring scale

3. Procedure

1. Take a **sample AAC block** and clean its surface.
2. Weigh the block and record the **initial weight (W_1)**.
3. Place the block in a **drying oven at 105–110 °C** for about **24 hours**.
4. Remove the block and allow it to **cool to room temperature**.
5. Weigh the block again and record the **dry weight (W_2)**.

4. Calculation

$$\text{Moisture Content (\%)} = \frac{W_1 - W_2}{W_2} \times 100$$

Where:

- **W_1 = Initial weight of block**
- **W_2 = Oven-dry weight of block**

5. Specifications

According to **Bureau of Indian Standards – IS 2185 (Part 3)**:

- Moisture content of AAC blocks should generally be **$\leq 10\%$ by weight** at the time of use.

6. Result

The **moisture content percentage** is calculated and compared with the **standard value** to determine whether the AAC blocks are suitable for construction.

Moisture Content Test on AAC Blocks



Weighing the Sample (Wet Weight)



Oven Drying Process



Moisture Content Calculation

$$\text{Moisture Content} = \frac{W_w - W_d}{W_d} \times 100$$


Wet Weight (Ww) **Dry Weight (Wd)**

Ww = Wet Weight **Wd = Dry Weight**

Dimensional Accuracy Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To check whether the **length, width, and height of AAC blocks** are within the specified limits and have proper shape for construction.

2. Apparatus

- Steel scale or measuring tape
- Vernier caliper
- Straight edge

3. Procedure

1. Select **5–10 AAC block samples** randomly.
2. Measure the **length, width, and height** of each block using a steel scale or vernier caliper.
3. Check the **edges, corners, and surface finish** of the blocks.
4. Compare the measured dimensions with the **standard size**.

4. Standard Size of AAC Blocks

Common standard size:

- **600 mm × 200 mm × 100 mm**
- **600 mm × 200 mm × 150 mm**
- **600 mm × 200 mm × 200 mm**

5. Specifications

- **Length tolerance:** ± 3 mm
- **Width tolerance:** ± 2 mm

- **Height tolerance:** ± 2 mm

6. Result

If the measured dimensions fall within the **permissible tolerance limits**, the AAC blocks are considered **dimensionally accurate and suitable for construction**.



Drying Shrinkage Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **amount of shrinkage (reduction in size)** of AAC blocks when moisture evaporates during drying.

2. Apparatus

- Vernier caliper / dial gauge
- Measuring scale
- Drying oven
- Water tank
- Comparator or measuring frame

3. Procedure

1. Take a **sample AAC block or prism specimen**.
2. Measure and record the **initial length of the specimen (L_1)**.
3. Immerse the specimen in **water for a specified period** to achieve saturation.
4. Remove the specimen and allow it to **dry under controlled conditions**.
5. Measure the **final length of the specimen after drying (L_2)**.

4. Calculation

Drying Shrinkage = $L_1 - L_2$

Unit: mm/m

Where:

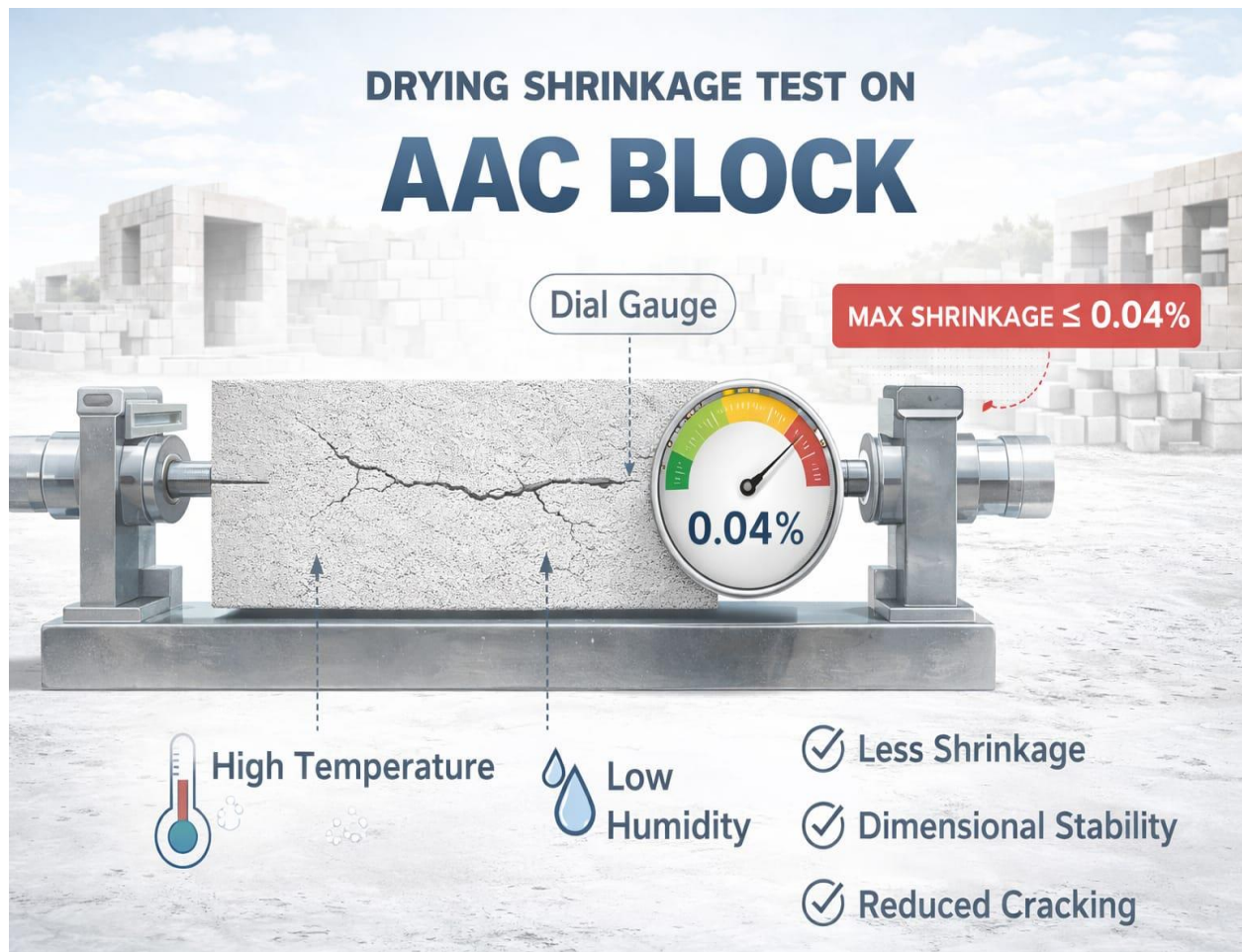
- L_1 = Initial length
- L_2 = Final length after drying

5. Specifications

- Maximum drying shrinkage of AAC blocks: **0.04% (0.4 mm per meter)**

6. Result

If the **drying shrinkage value is within the permissible limit**, the AAC blocks are considered **stable and suitable for construction**.



Thermal Conductivity Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **thermal conductivity (heat transfer ability)** of AAC blocks.
This test shows how well the block **insulates against heat**.

2. Apparatus

- Thermal conductivity apparatus (Heat flow meter / guarded hot plate)
- Temperature sensors (thermocouples)
- Power supply
- AAC block specimen

3. Procedure

1. Prepare a **dry AAC block specimen** of known dimensions.
2. Place the specimen between the **hot plate and cold plate** of the thermal conductivity apparatus.
3. Maintain a **constant temperature difference** between the two plates.
4. Allow heat to flow through the AAC block until **steady temperature conditions** are reached.
5. Measure the **heat flow and temperature difference** across the specimen.

Specifications

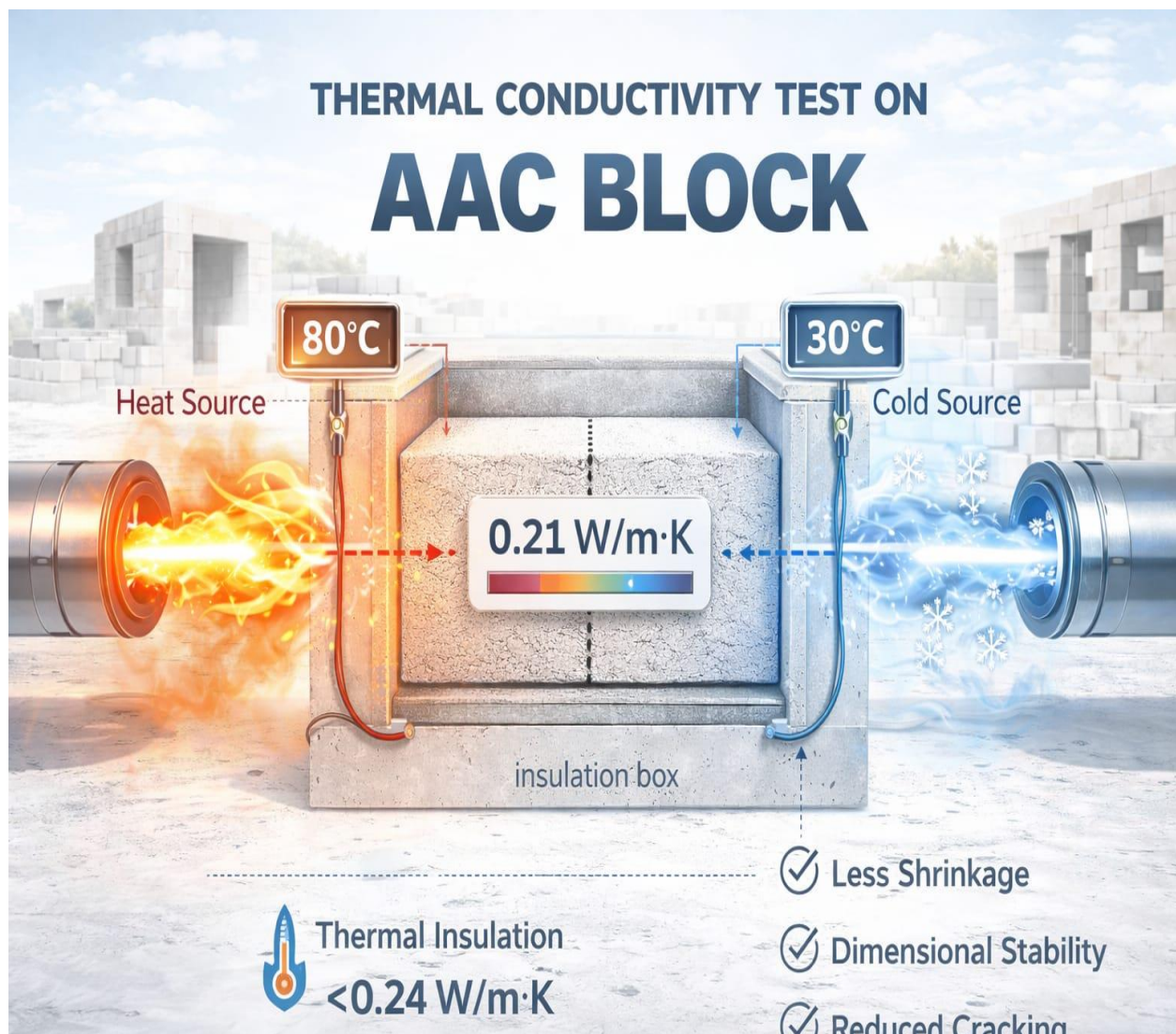
Typical thermal conductivity of AAC blocks:

0.16 – 0.24 W/m·K

This low value means AAC blocks provide **excellent thermal insulation**.

Result

The measured value of **thermal conductivity** indicates the **heat insulation capacity** of the AAC block used in construction.



Fire Resistance Test on Autoclaved Aerated Concrete (AAC) blocks

1. Objective

To determine the **ability of AAC blocks to withstand high temperatures and resist fire** for a specific period without losing strength or stability.

2. Apparatus

- Fire resistance furnace
- Temperature measuring devices (thermocouples)
- AAC block wall specimen
- Timing device

3. Procedure

1. Construct a **sample wall panel using AAC blocks**.
2. Place the specimen in a **fire resistance testing furnace**.
3. Expose one side of the wall to **high temperature according to standard fire conditions**.
4. Measure the **temperature rise on the opposite side** using thermocouples.
5. Record the **time until cracks, failure, or excessive heat transfer occurs**.

4. Observation

- Note the **time the AAC block wall resists fire without structural failure**.
- Observe **cracking, deformation, or temperature increase**

5. Specifications

AAC blocks have **excellent fire resistance** because of their air-pore structure.

Typical fire resistance:

- **100 mm thick wall:** about **2 hours**
- **150 mm thick wall:** about **3–4 hours**
- **200 mm thick wall:** about **4–6 hours**

6. Result

AAC blocks show **high fire resistance and good thermal protection**, making them suitable for **safe building construction**.

